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# SIGNALS, ILLUMINATION AIRCRAFT, SINGLE STAR AN-M43A2, AN-M4&A2, AND AN-M45A2 PARTS, AND LOADING, ASSEMBLING AND PACKING

*This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers the parts and the loading, assembling, and packing for one type of aircraft single star signals designated as follows:

AN-M43A2 - Red Star
AN-M44A2 - Yellow Star
AN-M45A2 - Green Star

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation - Packaging, Methods of
MIL-P-223 - Powder, Black
MIL-A-2550 - Ammunition, General specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).

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MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-105 - Inspection, Lots, Sampling, Examination, and Testing
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1234 - Pyrotechnics: Sampling, Inspection and Testing
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

*DRAWINGS

U.S. ARMY PICATINNY ARSENAL

PRODUCT

8847467 - Signal, Illumination, Aircraft, Single Star, AN-M43A2 thru AN-M45A2 Assemblies

PACKING

8836949 - Carton, Packing, Ammunition, For Signals, Illumination Aircraft, and Simulator Projectile, Air Burst
8836950 - Box, Packing, Ammunition, For Signals, Illumination, Aircraft, and Simulator Projectile, Air Burst.

INSPECTION EQUIPMENT

9201136 - Tunnels, Light
9201268 - Procedure, Light Output Measurement
9201390 - Procedure, Photocell Checkout
9247071 - Photocell

(Copies of Specifications, standards, and drawings required by suppliers in confection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

*For meaning of asterisk see 6.9.



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## 3. REQUIREMENTS

3.1 Material. - Materials and parts shall be In accordance with applicable drawings and specifications.

3.2 Assembly. - The assembly shall comply with all requirements specified on Drawing (dwg.) 8847467 and with all requirements specified in applicable specifications.

### 3.3 Moisture content

3.3.1 Star charge composition. - The moisture content of the red star charge composition at the loading station shall not exceed 0.1 percent, and the moisture content of the green and yellow star charge composition shall not exceed 0.5 percent.

3.3.2 Black powder. - The moisture content of the black powder, at the loading station at the time of loading the signals, shall not exceed 0.3 percent.

3.3.3 paper, and chipboard. - The moisture content of the paper and chipboard components, at the loading station at the time of loading the signals shall not exceed 6 percent.

3.3.4 Polypropylene felt. - The moisture content of the polypropylene felt components at the loading station at the time of loading shall not exceed 0.1 percent.

3.4 Transportation vibration. - The signal assembly shall comply with the following requirements:

3.4.1 The signal assembly shall not function during the test.

3.4.2 The signal assembly shall be safe to transport following the test (see 6.6).

3.4.3 There shall be no evidence of external damage to the signal assembly that will affect the intended function (see 6.7).

3.4.4 There shall be no evidence of damage in the packing or packaging that would affect the storage life of the signal assemblies.

3.4.5 The signal assembly shall comply with the requirements of 3.5 following this test.



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3.5 Functioning (Proving Ground). - The signals shall function in accordance with the following requirements:

3.5.1 No star shall burst in the pistol.

3.5.2 The signal case shall not split, rupture, or bulge to the extent that it sticks or “freezes” in the pistol barrel.

3.5.3 The primer shall not fail to function.

3.5.4 The star shall not fail to attain a minimum (min.) altitude of 150 feet above the firing point.

3.5.5 The star shall not fail to Ignite and reach approximately 1/4 of full brilliance within 2 seconds after firing.

3.5.6 Portions of composition that become detached during star burning shall not bum for more than two seconds.

3.6 Air leakage test. - The signal shall show no evidence of leakage.

3.7 workmanship. - All parts shall be fabricated and loaded in a through workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required) corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. (All packing components shall be dry).

3.8 First Article Inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified In the contract.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements



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as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define. terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted.

a. A statement that the lot complies with all of the quality assurance provisions specified In this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced In any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agents authority will not be required with subsequent certificates unless during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission. - Prior to the start of regular production the contractor shall submit a first article sample (see 6.1.c) to a Government approved facility as designated by the contracting officer for evaluation In accordance with the provislons Of' 4.2.2. The first article sample shall consist of the following quantities which have been produced by the contractor or funished by a supplier and which have been manufactured using



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the same production processes, Procedures and equipment which will be used in fulfilling the contract:

a. Fifteen (15) complete sets of Inert parts (i.e. 15 each of every inert component and Inert sub-assembly).
b. Ten (10) star assemblies of each color.
Thirty (30) signal assemblies of each color completely packaged and sealed as required by the applicable drawings.

All parts and materials, including packaging and packing components shall be obtained from the same source of supply as will be used in regular production. prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and he shall submit a record of this inspection with the sample, including statements of findings and certificates of conformance of materials. A sample containing known defects shall not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the Contracting Officer shall also be submitted whenever there Is a lapse In production for a period in excess of' 90 days, or whenever a change occurs In manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government. Unless otherwise directed by the contracting officer loaded components shall not be submitted for proving ground testing until all other phases of the first article samples have been accepted.

4.2.2 First article inspection provisions

4.2.2.1 Inert parts.

4.2.2.1.1 Submission. - The 15 sets of inert parts (see 4.2.la) shall be submitted to a Government approved laboratory designated by the contracting officer.

4.2.2.1.2 Inspections to be performed. - The inert parts will be subjected by the Government to any or all of the applicable examinations and tests specified in 4.3.2 and 4.3.3 of this specification and any or all of the requirements of the applicable drawings.

4.2.2.1.3 Rejection. - If any inert component or sub-assembly falls to comply with any or all of the applicable requirements, the first article samlpe shall be rejected. The Government reserves the right to terminate Its inspection upon any failure of an inert component or sub-assembly to comply with any of the stated requirements.



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4.2.2.2 Star assemblies

4.2.2.2.1 Submission. - The 10 star assemblies (ace 4.2.1b) shall be submitted to a Government approved laboratory designated by the contracting officer.

4.2.2.2.2 Inspections to be performed. - The star assemblies, shall be subjected to the static test for candlepower, color value and burning time in accordance with the method specified in 4.4.5.

4.2.2.2.3 Rejection. - The first article sample shall be rejected if any star assembly falls to comply with any of the requirements as classified in 4.3.3.5, Table IV. The Goverment reserves the right to terminate its Inspection upon encountering any failure of an item in the sample to comply with any or all requirements.

4.2.2.3 Loaded signal assembles

4.2.2.3.1 Submission. - The 30 loaded signal assemblies (see 4.2.1c) shall be submitted to a Government proving ground designated by the contracting of'fleer.

4.2.2.3.2 Inspections to be performed. - Prior to submission of the signal assemblies to the proving ground for functioning test, the loaded components shall be subjected to the following tests at the contractor's plant by the Government:

a. Transportation vibration (see 3.4) - 30 signal assemblies of each designation.
b. Air leakage (see 3.6) - 30 signal assemblies of one designation selected from the assemblies after test specified in "a" above.

After completion of' the transportation vibration and air leakage tests, each signal assembly shall be conditioned and fired at the proving ground as specified in 4.4.3.

4.2.2.3.3 Rejection. - The first article sample shall be rejected if any loaded signal assembly fails to comply with:

a. The requirements of 3.4 as classified In 4.3.3.2, Table II
b. The requirements of 3.6 as classified in 4.3.3.4
The requirements of 3.5 as classified In 4.3.3.3, Table III.



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The Government reserves the right to terminate its inspection upon encountering any failure of an Item in the sample to comply with any or all requirements.

4.3 Inspection provisions

4.3.1 Lot formation. - The term “inspection lot” as used In this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior to subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the Inspection lot has been produced by one manufacturer In one unchanged process, using the same materials and methods in accordance with the sane drawings, same drawing revisions, same specifications, and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of and inspection lot . All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type Of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded signals shall contain:



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for major and minor defects shall be in accordance with MIL-STD-105 Inspection Level II except that continuous sampling plans In accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method Is elected, It shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Case, signal (ace dug. 8847475 covering a detail of dwg. 8847467).

| Categories | Defects | Method of Inspection | Code No. (see 6.2) |
|---|---|---|---|
| Critical: | None defined | | |
| Major: | AQL 0.40 percent | | |
| 101. | Diameter of primer hole, maximum, (max) | Gage | 01001 |
| Minor: | AQL 4.00 percent | | |
| 201. | Inside diameter, max. | Gage | 01002 |
| 202. | Outside diameter, min. | Gage | 01003 |
| 203. | Diameter of flange | Gage | 01004 |
| 204. | Angle of flange | Gage | 01005 |
| 205. | Thickness of base at primer hole area | Gage | 01006 |
| 206. | Thickness of base in area above propelling charge cavity | Gage | 01007 |
| 207. | Radii missing | Visual | 01008 |
| 208. | Finish Improper | Visual | 01010 |
| 209. | Evidence of poor workmanship (see 3.7) | Visual | 01009 |

l 4.3.2.2 Signal case assembly (see dwg. 8847468 covering a detail of dug. 8847467

| Categories | Defects | Method Of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined | | |



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| | | | |
|---|---|---|---|
| Major: | AQL 0.40 percent | | |
| 101. | Primer above flush or more than max. below flush . . . . . | Gage | 13001 |
| 102. | Primer not sealed 360° . . . . . | Visual | 13002 |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship (see 3.7) | Visual | 13003 |

4.3.2.3 Cup (see dwg. 8847464 or 9240782 when applicable covering a detail of dwg. 8847467).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined | | |
| **Major:** | AQL 0.40 percent | | |
| 101. | Inside diameter | Gage | 14001 |
| 102. | Thickness of wall | Gage | 14002 |
| Minor: | AQL 1.00 percent | | |
| 201. | Diameter of small holes . . . . . | Gage | 14003 |
| 202. | Diameter of' bottom hole (when applicable ) | Gage | 14004 |
| 203. | Evidence of poor workmanship (see 3.7) | Visual | 14005 |

4.3.2.4 Star assembly (see dug. 8847463 covering a detail of dwg. 8847467).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | | | |
| 1. | Color identification incorrect . . . | Visual | 02001 |
| Major: | AQL 0.65 percent | | |
| 101. | Length of protruding quickmatch. | Gage | 02002 |
| 102. | Priming charge distribution of total hole area, disc and bottom, less than 2/3 min. . . . . | Visual | 02003 |
| 103. | Quickmatch missing . . . . . | visual | 02004 |
| 104. | Disc loose . . . . . | Manua1 | 02005 |



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| | | | |
|---|---|---|---|
| Minor: | AQL 1.00 percent | | |
| 201. | Crimp not full 360 degrees ..... | Visual | 02007 |
| 202. | Evidence of poor workmanship (see 3.7) . . . . . | Visual | 02006 |

4.3.2.5 Top (see dwg. 8847473 covering a detail of dwg. 8847467).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined | | |
| Major: | AQL 0.40 percent | | |
| 101. | Crack or split . . . . . | Visual | 03001 |
| Minor: | AQL 2.50 percent | | |
| 201. | Thickness . . . . . | Gage | 03002 |
| 202. | Outside diameter at datum height. | Gage | 03003 |
| 203. | Protrusion of sides above bottom. | Gage | 03004 |
| 204. | Length of sides . . . . . | Gage | 03005 |
| 205. | Bare spot in protective coating or coating missing ..... | Visual | 03006 |
| 206. | Evidence of poor workmanship (see 3.7) . . . . . | Visual | 03007 |

*4.3.2.6 Assembly, prior to inserting star assembly retaining washer, and wad-disc assembly (see dwg. 8847467).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | | | |
| 1. | Propelling charge missing . . . . . | Visual | 04001 |
| 2. | Propelling charge weight less than min. (see note) . . . . . | Balance | 04004 |
| Major: | None defined | | |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship (see 3.7) . . . . . | Visual | 04003 |

NOTE : Volumetric weighing may be utilized as a substitute for precise weighing following qualification. Qualification shall consist of' precise weighing until such time as twenty-five hundred



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(2,500) consecutively volumetric loaded charges are found to meet the minimum prescribed weight. After successful completion of the qualification quantity, the contractor may use a 32-O-1 check weighing sampling plan for each hours production of powder loaded signal cases. If a *'weight under min." critical defect is found In the thirty-two (32) unit sample, the hours production represented by the sample plus enough additional units to total eight hundred and forty (84o) consecutive, defect free units, must be produced and check weighed before returning to the 32-O-1 hourly sampling plan.

**4.3.2.7 Assembly, prior to inserting star assembly (see dwg. 8847467).**

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| **Critical:** | | | |
| 1. | Wad-disc assembly missing, not seated, or disc punctured .*..... | Visual/ Manual | 05001 |
| Iuq or : | None defined | | |
| **Minor:** | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship (see 3.7) . . . . . | Visual | 05002 |

**4.3.2.8** Assembly, prior to Inserting top (see dwg. ***8847467).***

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| **Critical:** | | | |
| 1. | Star assembly color incorrect . . . . . | Visual | 06001 |
| MaJor: | AQL 0.40 percent | | |
| 101. | Star assembly missing, inverted, or damaged to extent that function may be impaired . . . . . | Visual | 06002 |
| 102. | Spacing component or wad missing . . . . . | Visual | 06003 |
| **Minor** | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship (see 3.7) .** .. .... ..... | Visual | 06004 |



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4.3.2.9 Signal assembly (see dwg. 8847467).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | | | |
| 1. | Band color Incorrect | Visual | 07001 |
| Major: | AQL 1.00 percent | | |
| 101. | Top insecure or not fully seated | Manual | 07002 |
| 102. | Contents loose | Manual | 07003 |
| 103. | Color band missing | Visual | |
| 104. | Marking misleading or unidentifiable | Visual | 07005 |
| 105. | Case split or assembly otherwise damaged | Visual | 07006 |
| 106. | Sealing compound missing from mating surfaces of top and signal case | Visual | 07007 |
| Minor: | AQL 1.50 percent | | |
| 201. | Total length | Gage | 07008 |
| 202. | Bare spot in protective coating | Visual | 07009 |
| 203. | Finish improper | Visual | 07010 |
| 204. | Evidence of poor workmanship (see 3.7) | Visual | 07011 |

*4.3.2.10 Carton, prior to sealing (see dug. 8836949).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined | | | |
| Major: | AQL 0.40 percent | | |
| 101. | Number of assemblies in carton incorrect | Visual | 08001 |
| 102. | Assembly Improperly packed In carton (inverted or packing material missing) | Visual | 08002 |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship (see 3.7) | Visual | 08003 |



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*4.3.2.11 Sealed carton (see dwg. 8836949).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | | | |
| 1. | Color identification missing or incorrect | Visual | 09001 |
| Major: | AQL 0.40 percent | | |
| 101. | Sealing strip improperly applied. | Visual | 09002 |
| 102. | Carton damaged to extent that contents are exposed or liable to become exposed | Visual | 09003 |
| Minor: | AQL 1.50 percent | | |
| 201. | Contents loose | Manual | 09004 |
| 202. | Marking misleading or unidentifiable | Visual | 09005 |
| 203. | Evidence of poor workmanship (see 3.7) | Visual | 09006 |

* 4.3.2.12 Wood packing box, prior to sealing (see dwg. 8836950).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined | | |
| Major | AQL 0.40 percent | | |
| 101. | Carton inverted (primer end up) | Visual | 10001 |
| Minor: | AQL 1.00 percent | | |
| 201. | Incorrect number of cartons | Visual | 10002 |
| 202. | Evidence of poor workmanship (see 3.7) | Visual | 10003 |

l 4.3.2.13 Sealed wood packing box (see dwg. 8836950).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | | | |
| 1. | Color identification missing or incorrect | Visual | 11001 |



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| | | | |
|---|---|---|---|
| Major: | AQL 0.65 percent | | |
| 101. | Box damaged to extent that contents are exposed or liable to become exposed . . . . . | Visual | 11002 |
| 102. | Hardware or strapping missing broken or loose . . . . . | Visual/ Manual | 11003 |
| 103. | DOD symbol missing or unidentifiable ..... | Visual | 11004 |
| Minor: | AQL 1.50 percent | | |
| 201. | Hardware or strapping, improperly engaged or assembled . . . . | Visual/ Manual | 11005 |
| 202. | Metallic seal missing, unsealed, or Improperly positioned . . . . . | Visual | 11006 |
| 203. | Contents loose . . . . . | Manual | 11007 |
| 204. | Marking missing, misleading, or unidentifiable . . . . . | Visual | 11008 |
| 205. | Evidence of poor workmanship (see 3.7) ..... | Visual | 11009 |

4.3.2.14 Barrier Bag, Sealed (see dwg. 8836949).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined | | |
| Major: | AQL 0.40 percent | | |
| 101. | Seal improper or incomplete . . . | Visual | 16001 |
| 102. | Bag punctured, torn or cut . . . . | Visual | 16002 |
| Minor: | AQL 1.00 percent | | |
| 201. | Marking missing, Incorrect or unidentifiable . . . . . | Visual | 16003 |
| 202. | Evidence of poor workmanship (see 3.7) ..... | Visual | 16004 |

4.3.2.15 Workmanship. - Applies to all components and assemblies not specifically listed In 4.3.2.1 through 4.3.2.14 (see applicable drawings).

| Categories | Defects | Method of Inspection |
|---|---|---|
| Critical: | None defined | |
| Major: | AQL 0.40 percent per defect | |
| 101. | Operation missing or improperly perfomed . . . . . | Visual |



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102. Burr or chip on or at intersection or other functional surfaces . . . . . Visual/Manual
103. Finish of sliding or other functional surface improper ... Visual
104. Component missing or improperly assembled . . . . . Visual
105. Protective finish missing or contains bare spot . . . . . visual

Minor; AQL 0.65 percent
201. Evidence of poor workmanship other than that covered under Major category above (see 3.7). Visual

4.3.3 Production Testing

4.3.3.1 Moisture content (see Table I) Major defects

TABLE I

| Material | Code No. |
|---|---|
| Star charge composition (see 3.3.1) | 12001 |
| Black powder (see 3.3.2) | 12002 |
| Felt (see 3.3.4) | 12003 |
| Paper (see 3.3.3) | |
| Chipboard (see 3.3.3) | 12005 |

The contractor shall provide adequate controls to insure that the materials comply with the requirements. For verification, the contractor shall select and test one sample of each material used in each eight hours production. A composite sample shall not be used. If the moisture content of a sample exceeds the requirement and loadlng has not begun, the material represented by the sample shall be rejected. If components have been loaded with material containing excessive moisture, the loaded components and the remaining unloaded material shall be rejected. Test procedures shall be as specified in 4.4.1. The sample of star charge composition shall be selected prior to consolidating the star assemblies.

4.3.3.2 Transporation vibration.- The signal assemblies shall be observed and examined visually without disassembly for any evidence of failure to comply with the requirements as classified in Table II.



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TABLE II

| Defect | Classification | Code No. |
|---|---|---|
| Signal functions during test (see 3.4.1) | Critical | 07012 |
| Signal not safe to transport following test (see 3.4.2) | Critical | 07013 |
| Signal damaged after test (see 3.4.3) | Major | 07014 |
| Signal packing damaged (see 3.4.4) | Major | 07015 |

4.3.3.2.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified, two hundred and forty (240) signal assemblies shall be selected from each lot for test. The lot shall be rejected if any defect as classified in Table II occurs. The test shall be performed as specified in 4.4.2.

4.3.3.2.2 After three (3) consecutive lots have complied with the acceptance criteria of 4.3.3.2.1, eighty (80)signal assemblies shall be selected from each lot for test. The lot shall be rejected if any defect as classified In Table II occurs.

*4.3.3.3 Functioning. - The signal assemblies shall be observed for any evidence of failure to comply with the requirements as classified in Table 111 when tested as specified In 4.4.3.

TABLE III

| Defect | Classification | Code No. |
|---|---|---|
| Star color incorrect (see dwg. 3847463) | Critical | 07017 |
| Star bursts In pistol (see 3.5.1) | Critical | 07018 |
| Star burning time below min. (see dwg. 8847463) | Major | 07019 |
| Altitude below min. (see 3.5.4) | Major | 07020 |
| Primer fails to function (see 3.5.3) | Major | 07021 |



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| | | |
|---|---|---|
| Star intensity and Ignition time over max. (see 3.5.5) | Major | 07022 |
| Detached composition burning time over max. (see 3.5.6) | Major | 07023 |
| Case sticks or freezes in barrel (see 3.5.2) | Major | 07024 |
| Star fails to ignite (see 3.5.5) | Major | 07027 |

4.3. 3.3.1 The two hundred forty (240) signal assemblies selected in 4.3.3.2.1 shall be forwarded to the Proving Ground for this test. The lot shall be rejected if a Critical defect occurs or if eight (8) or more Major defective are found (see Table III).

4.3. 3.3.2 The eighty (80) signal assemblies selected in 4.3. 3.2.2 shall be forwarded to the Proving Ground for this test . The lot shall be rejected If a Critical defect occurs or if four (4) or more Major defectives are found (see Table III).

4.3.3,4 Air leakage test (see 3.6) - Major defect - Code No. 07025. - This test shall be performed 100 percent in accordance with the procedure specified in 4.4.4. All assemblies that fall to comply with the requirement shall be classed defective and removed from the lot.

4.3.3.5 Static test of star assembly (see dwg. 8847463). - The star assemblies shall be tested and observed for the defects as classified in Table IV.

TABLE IV

| Defect | Classification | Code No. |
|---|---|---|
| Assembly falls to ignite | Major | 02008 |
| Candlepower under min. | Major | 02009 |
| Color value under min. | Major | 02010 |

4.3.3.5.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the l acceptance criteria specified, eighty (80) star assemblies of each color shall be selected from each lot for test in accordance with the procedure specified in 4.4.5. The lot shall be rejected if five (5) or more defective are found.

4.3.3.5.2 After three consecutive lots have complied with the acceptance criteria of 4.3.3.5.1, thrity-two (32) star as-



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semblies of each color shall be selected from each lot for test. The lot shall be rejected, If, during the test, three (3) or more assemblies exhibit any of the defects as listed in Table IV.

4.3.3.6 Chamber gaging (see dwg. 8847467) Major defect - Code No. 07026. - Chamber gaging shall be performed 100 percent. Any assembly that binds, or otherwise falls to gage shall be classed defective and removed from the lot. The gaging shall be performed as specified in 4.4.6.

4.3.3.7 Check test for possible deterioration of primers (see applicable primer specification). If the total elapsed time between original acceptance of any primer lot and the assembly of that lot into the signal assemblies exceeds two years, or if the primers have been subjected to adverse conditions, however brief at any time since previous tests, the primer lots shall be subjected to and must satisfactorily pass the check test specified in the applicable primer specification. The check test shall be performed by the contractor (see 6.5) prior to assembling the primers into the signals.

4.3.3.8 Heat seal test of sealed bag. - (see dwg. 8836949) Major defect, Code No. 15001. Sampling acceptance criteria, and test methods shall be as specified in MIL-P-116.

4.3.4 Inspection Equipment .-Inspection equipment required to perform the inspections prescribed in this specification is identified in the Examination (4.3.2) and Test Method (4.4) paragraphs herein. See 6.3 for details concerning responsi-bilities for inspection design and approval.

4.4 Test methods and procedures.

4.4.1 Moisture content

4.4.1.1 Star charge composition

4.4.1.1.1 preferred Method. - The Karl Fischer method as stated in MIL-STD-1234, Method 101.2 up to paragraph 5.3 shall be used. A fifty gram (gin) sample plus or minus 0.1 gm. shall be added to a 500 millilter (ml) volumetric flask containing approximately 300 to 400 ml. of methanol and 25 gins. Of dry sodium nit.rate. The flask shall be stoppered and the contents swirled cautiously for several minutes until the material is thoroughly dispersed The sample shall be allowed to remain in contact with the methanol for approximately two hours, then the 500 ml. volumetric flask shall be filled up to the 500 ml. mark with methanol and swirled again. A blank without the sample shall be put through the same procedure . after the sample has settled a 50 ml. aliquot of the clear supernatant ant liquid shall be withdrawl and immediately transferred to the standard titraticn



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vessel containing approximately 100 ml. of methanol which has just been titrated to the prelimimary end point as described In MIL-STD-1234, Method 101.2 paragraph 5.1. The final end point shall be reached in 3.5 minutes In the manner described in MIL-STD-1234, Method 101.2, paragraph 5.4. A 50 ml. aliquot of the blank shall be titrated in the same manner. The water content shall be calculated as follows:

Percent water = (VR-S) (V'R-S'). Multiply the result by 100F and divide by W.

where:

F = gins. of water per ml. of standard water in methanol solution.
V = ml. of Karl Fischer reagent added to the sample
V'= ml. of Karl Fischer reagent added to the blank
R = ml. of standard water In methanol solution per ml. of Karl Fischer reagent.
S = ml. of standard water in methanol solutlon for titration of sample.
S'= ml. of standard water in methanol solution for back titration of blank.
W = weight of sample In gins.

4.4.1.1.2 Alternate method. - The moisture content shall be determined In accordance with Method T101.4 of MIL-STD-286.

4.4.1.2 Black powder. - The method of determining the percentage of moisture in black powder shall be in accordance with MIL-P-223.

4.4.1.3 Polypropylene felt, paper, and chipboard. - Ten grams of each material shall be accurately weighed and placed in a tared weighing dish. The dish and contents shall be weighed and placed in an oven and dried at 100 plus or minus 2 degrees Centigrade for 2 hours. The dish shall be cooled in a desiccator and weighed. The loss in weight shall be calculated as percent moisture in the sample.

4.4.2 Transportation vibration. - The signal assemblies selected as specified in 4.3.3.2.1 or 4.3:3.2.2 shall be packaged and packed in accordance with dwgs. 8836949 and 8836950. Each packed box shall be subjected to the transportation-vibration test as specified in MIL-STD-331, except that each box shall be vibrated at the specified amplitudes for four (4) hours in each of three different positions (i.e., box positioned so that



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signal assemblies are vertical with base end down, box positioned so that signal assemblies are horizontal, and box positioned so that the signal assemblies are vertical with base end up. After the test, the box packing and the signal assemblies shall be examined to determine compliance with the requirements.

4.4.3 Functioning (Proving Ground). - The signals shall be immersed to a depth between 6 and 9 inches for 2 hours in water maintained at 70 plus or minus 10 degrees Fahrenheit, At the end of the immersion time, the signals shall be removed from the water and the exterior surfaces wiped dry. The signals shall be fired in a pistol, for which the signal is standard, that is mounted on a tower platform at a min. height of 100 feet above the ground. The signal functioning shall be observed for compliance with the requirements. Test shall be conducted in accordance with Acceptance Test Procedure SI-A-VAR Suppl. - SI-A-1

4.4.3.1 Test Validity. - If for any reason the proving grcund considered that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

*4.4.4 Air leakage test. - The signal assembly shall be placed in a cylindrical air tight chamber with the free space minimized. The air pressure within the chamber shall be raised to a minimum of 3 pounds p.s.i.g., by means of a fixed volume of air. With the air supply shut. off the pressure shall be maintained for 15 seconds min. An accurate pressure measuring instrument shall be observed for evidence of' signal leakage.

*4*4,5 Static test of star assembly. - The star assemblies shall be tested by supporting the star In a horizontal position with the axis of the star perpendicular to the photometric axis, and ignited by means of quickmat cl.. The candlepower and color value shall be measured and recorded to determine compliance with the candlepower and color value requirements specified on the applicable drawing Determination of candlepower and color value shall be made in accoradance with the procedures and equipmemt specifled in dwgs. 9201136, 9201260, 9201390 and associated dwgs., 9201392 and 9247071.

4.4.6 Chamber gaging - The signal assembly shall be gaged with equipment specified in 4.3.4.

5. PREPRATION FOR DELIVERY

5.1 Preservation and Packaging.

5.1.1 Level A. - Signals shall be packaged in accordance with dwg. 8836949.

5.2 Packing

5.2.1 Signals shall be packed in accordance with dwg. 8836950.



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5.3 Marking. - Marking shall be In accordance with dwgs. 8836949 and 8836950.

*6 NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified In Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers .-The five digit cede numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. They should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3).

6.3 Inspection Equipment Designs .-Inspection equipment designs are or two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.
b. A source control drawing or a specification control drawing as defined In MIL-STD-100.
c. An envelope drawing, as defined In MIL-STD-100, meet. When envelope drawings are specified the contractor shall prepare designs which comply with the criteria therein.



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6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating Instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five digit defect code number (or other specific identifying Information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval .-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract The contractor must include in his submission of designs a listing of any and all Government SIE designs which will be utilized. Partial submission of inspection designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Star burning time. - The star burning time is defined as the time from the appearance of the first colored light from the illuminant assembly (not the first sputtering light from the quickmatch or the black powder priming charge until the time that the light from the star has decreased to approximately less than 10 percent of peak brilliance.



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6.5 Cost of check test. - The contracting officer should arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test. The tests shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the primers in the signals and shall not constitute a basis for rejection against either contractor except where deterioration has occured as a direct result of carelessness in handling, storage, etc., permitted while the primer lots were under the jurisdiction of either contractor.

6.6 The signal assembly shall be considered safe to transport provided no evidence exists of loose powder, or composition in the box, or missing, loose, or protruding primers.

6.7 The signal assemblies will be considered free of damage that will affect the intended function provided the top seal has not been broken by movement or displacement of the top, or the case side has not been distorted sufficiently to prevent ejection of the star assembly.

6.8 Identification of international standardization agreements. - Certain provisions of this specification are the subject of International standardization agreement STANAG 3398-Air Pyrotechnics Interchangeability Chart. When amendment, revision or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

6.9 The margins of this specification are marked with an asterisk to Indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any Inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.10 Proving ground test summary:



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| Lot | Sample Size | Test | Requirements |
|---|---|---|---|
| First Article | 30 of each color | Functioning | See Table III |
| Regular Production (First three consecutive acceptable lots) | 240 | Functioning | See Table III |
| Remaining | 80 | Functioning | See Table III |

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1370-A-032



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No 22-R255

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers, vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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